

Chapter 1

From Empirical Percentiles to Hypothesis Testing: Validating the Selection of Least-Contaminated Reference Sites

1.1 Introduction

Sediment contamination poses chronic threats to benthic communities in aquatic ecosystems, and the assessment of chemical stress requires careful selection of ecologically relevant variables. The 16 chemical variables retained in this study were chosen by ecologists and toxicologists and justified from an ecological and toxicological standpoint, reflecting the judgement that each variable carries meaningful stress information for the communities under study.

Despite their distinct contamination roles, these 16 variables do not behave as equally distinct statistical entities. As random variables measured across sampling sites, many of them exhibit strong co-variation in concentration, reflecting shared contamination sources, co-deposition pathways, or common geochemical associations, and some show highly similar distributional patterns across sites. This statistical redundancy means that treating the 16 raw chemical measurements as independent lines of evidence

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would be misleading: correlated variables would carry overlapping information and receive collective weight, since the variance of their combined contribution exceeds the sum of their individual variances once their covariance is positive ¹. As a result, the underlying structure of contamination would be obscured rather than clarified. This raises a central methodological question: how can we extract a smaller set of latent stressors that are statistically orthogonal to one another while faithfully preserving the original contamination stress information? The value of identifying such latent stressors is: orthogonality ensures that any contamination stress signal attributed to one latent stressor is uncorrelated with all others. Then the contamination information retained within each latent stressor refers exclusively to its own domain, without carrying confounded information from the others.

With these orthogonal latent stressors, we adopt a specific contamination–community response perspective to synthesise stressor values into an integrated stress level for sites. This integrated stress level identifies sites whose communities are least affected by chemical contamination. Here it is important to distinguish two related but non-equivalent notions. A natural site refers to a community shaped under undisturbed or minimally disturbed conditions — a theoretical reference state representing biological integrity in the absence of human stressors. A least-polluted site, by contrast, is an operational concept defined relative to the observed distribution of integrated chemical stress within the study area, with sites at the low end of that distribution taken as the least-polluted.

We argue that, in this study, the least-polluted-site concept is the safer and indeed the only defensible choice of the author’s baseline. As studies (Stoddard et al., 2006 [14], Davies and Jackson, 2006 [5]) emphasise in developing the Biological Condition Gradient, truly natural conditions are difficult to observe directly: in extensively altered regions, practitioners tend to accept the "best of what remains" as the apparent optimum, because the original natural state can no longer be visualised. Moreover, they note that least-disturbed reference sites differ markedly across regions in their degree of departure from historical or natural condition. Direct observation of a nat-

¹For two recorded measures treated as random variables X and Y , the variance of their sum is $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$. When $\text{Cov}(X, Y) > 0$, this exceeds $\text{Var}(X) + \text{Var}(Y)$, so the shared covariance is absorbed into the combined variance and cannot be attributed to X or Y individually.

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ural community is therefore not attainable from the available data; the least-polluted sites, identified from the low tail of the integrated stress distribution, serve as the available approximation to natural community organisation.

The ecological significance of these least-contaminated sites follows directly. Within them, community structure is expected to be governed primarily by habitat and environmental factors rather than by chemical stress, so they retain the information of naturally organised communities. These natural community information are comparable benchmark — both for understanding natural community composition and for studying the community changes induced by contamination stress.

The central contribution of this chapter is methodological. Rather than delineating the least-contaminated group by a fixed empirical percentile—a convention widely adopted—we defend the delineation threshold as a question to be tested. Using two complementary resampling-based hypothesis tests, we ask whether a candidate reference group of a given size genuinely carries weaker contamination-constrained community variation than would be expected by chance, and we use this evidence to support a selected least-contaminated set rather than empirically assuming it. By generalizing this approach to different study cases, it provides informative guidance that least-contaminated sites need to be set according to the observed contamination stress and not by a fixed empirical percentile.

1.2 Methods

1.2.1 Principle Component Analysis/R-model Ordination on Chemical Matrix

In linear algebra, the eigen-decomposition of a matrix identifies a set of characteristic directions along which a linear transformation acts purely by scaling, and within a measure space it resolves the structure of that space into mutually orthogonal axes whose orthogonality can be formally proven from the symmetry of the underlying matrix (Strang, 2006 [15]). When this decomposition is applied to a data matrix, it acquires a statistical interpretation: Principal Component Analysis is an eigen-

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decomposition performed after a descriptive space — conventionally the column space — has been chosen for the data matrix, and in numerical ecology it is classified among the R-mode ordination methods that operate under a Euclidean distance measure of dissimilarity (Borcard, Gillet, and Legendre, 2018 [4]). PCA extracts the eigenvectors from the correlation matrix of the column space, and the resulting orthogonal eigenvectors are scaled so as to preserve, as far as possible, the structure and magnitude of the data within the original descriptive space. The values of the original variables are then projected onto these orthogonal vectors and come to lie in a new descriptive space spanned by the ordination axes, the principal components. It must be emphasised that orthogonality holds only among the newly constructed principal components; the covariance among the original variables is still partly retained within these components. This is reflected in the fact that the variance of each principal component preserves not only the original quantity of information (variance) but also the covariance information among the original variables.

In this study, the structure and magnitude of the variance carried by the principal components themselves are not our central concern. What concerns us is rather the assurance that the principal components preserve the original information content together with their mutual orthogonality, that is, their statistical independence. In this way, when we discuss the contamination stress information of one or several stressors at a given site, we can ensure that it is uncorrelated and completely retained within the domain to which the latent stressor(s) refer, allowing the contamination variation arising from different latent stressors to be correctly distinguished.

A brief proof of the mathematical properties underlying PCA is provided in the Appendix 1.5.1). Performing PCA on the chemical matrix, rotating the loadings, flipping to ensure most positive loadings across chemicals, and selecting the leading principal components, yields a set of latent stressors that quantify the contamination stress information. The full procedure, from the input chemical matrix to the output latent stressors, is summarised in Algorithm 1.

Algorithm 1: Construction of the PC-based chemical stressor matrix

Data: Site-by-chemical matrix $\mathbf{M}_{310 \times 16}$.

Result: PC-based chemical stressor matrix $\mathbf{S}_{310 \times p}$.

- 1 Log-transform and standardise, then form the correlation matrix:

$$\mathbf{M}_{310 \times 16} \rightarrow \mathbf{M}'_{310 \times 16} \xrightarrow{\text{correlation}} \mathbf{cor}_{16 \times 16}$$

- 2 Apply PCA, quartimax rotation (flip if needed), and loading selection:

$$\mathbf{cor}_{16 \times 16} \rightarrow \mathbf{Loadings}_{16 \times p'} \xrightarrow{\text{rotation}} \mathbf{Loadings}^*_{16 \times p'} \xrightarrow{\text{selection}} \mathbf{Loadings}^*_{16 \times p}$$

- 3 Project the processed chemical matrix onto the selected rotated loading

$$\text{space: } \mathbf{S}_{310 \times p} = \mathbf{M}'_{310 \times 16} \mathbf{Loadings}^*_{16 \times p}$$

- 4 **return** $\mathbf{S}_{310 \times p}$
-

1.2.2 Latent Stressors and Contamination Cumulative Impact

In multiple-stressor ecology, stressors are commonly defined externally as ecologically interpretable pressures, such as nutrient enrichment, habitat degradation, flow alteration, or pollution. Studies such as those of Halpern et al. and Birk et al. adopt this perspective to examine how multiple recognisable environmental pressures combine, dominate, or interact in shaping biological responses (Halpern et al., 2008 [8]; Birk et al., 2020 [3]).

In the present study, by contrast, the chemical stressors are not directly defined pressure categories. They are latent contamination gradients extracted through the procedure of the preceding subsection, and are best interpreted as contamination syndromes (Huang et al. [9] and Loska and Wiechuła [11]). The latent stressors here therefore do not denote individual pollutants, but rather structured chemical pressures potentially experienced by the benthic communities.

From these latent stressors, an integrated contamination stress is derived for each site. We adopt the perspective that the community-level effects of the latent stressors are *additive*: their individual stress contributions are summed to yield a single integrated contamination stress. This quantity is denoted the Sum Relative Stress (SumRel), which expresses the position of a site relative to all other sites within the summed-

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stressor value space. Accordingly, the SumRel of site i is the sum of its stressor values across the p latent stressors. These values are the raw principal-component scores of the log-transformed, standardised chemical data $\mathbf{M}'$ projected onto the retained loadings, sign-aligned to a common direction in which higher scores indicate higher chemical concentrations.

$$\text{SumRel}_i = \sum_{j=1}^p s_{ij}, \quad s_{ij} \in \mathbf{S}_{310 \times p},$$

where s_{ij} is the direction-aligned raw stressor value of site i on latent stressor j . Different contamination–community response perspectives may legitimately coexist. The additive view adopted here assumes that summing orthogonal stress values reflects the combined influence of the stressors on community composition (Halpern et al., 2008 [8]), whereas a dominance view holds instead that a single prevailing stressor obscures the others and governs the response, so that the integrated effect is set by the highest stressor value alone (Birk et al., 2020 [3]). We construct the integrated stress measure (SumRel) under the additive perspective, while acknowledging that the choice among such perspectives is one that ultimately be left to ecologists to judge.

1.2.3 Least-Contaminated Sites Selection and Validation

Proceeding from the integrated stress level, we define the *least-contaminated sites* as those lying at the low tail of the observed distribution of integrated contamination stress (SumRel). These sites constitute the least-contaminated group, whose taxa communities are expected to be the least affected by contamination and thus to serve as a benchmark, or baseline.

Determining this group generally requires a threshold to separate the least-contaminated sites from the remainder. In the study of Zhang (Zhang, 2008 [18]), this threshold was set at the 20th percentile of integrated stress. Such a 20% cut-off is, in most cases, chosen empirically (Australian and New Zealand Governments, 2018 [2]; Ministry for the Environment [12]). While this approach does capture the sites at the low end of the observed stress distribution, it neither examines how a fixed-percentile choice

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(20%) weakens the least-contamination assumptions underlying the delineated group size when applied across differing contamination settings (Stoddard et al., 2006 [14]), nor verifies that a low-percentile group of a given size exerts a significantly weaker influence on community composition than other groups of the same size.

Prescribing an absolute count threshold (for instance, 50 or 60 sites) applicable across all contamination settings is likewise neither attainable nor interpretable, since such a number depends on the distribution of stress levels in the study context and on the researcher’s judgement of what constitutes, and how to constitute, an ecologically “low” level of contamination.

1.2.4 Between-group resampling test

In this section we continue to adopt the 20th percentile to select the least-contaminated sites, but design a hypothesis test, based on a multivariate-regression resampling procedure, to verify whether the level of contamination impact on the community within the least-contaminated group is significantly lower than that of other, randomly drawn groups. The method draws on the first half of Redundancy Analysis (RDA), measuring the magnitude of contamination impact on the community within each group by fitting a multivariate regression model.

Specifically, for a given set of m sites with retained latent-stressor matrix $\mathbf{X}$ and community composition matrix $\mathbf{Y}$, we regress each species record $\mathbf{y}$ in $\mathbf{Y}$ on $\mathbf{X}$. The fitted records of all species are collected into $\hat{\mathbf{Y}}$, a matrix of the same dimensions as $\mathbf{Y}$. The variation in $\hat{\mathbf{Y}}$ is constrained by $\mathbf{X}$ and is taken as a measure of the community variation induced by the latent stressors that $\mathbf{X}$ represents. The adjusted coefficient of determination $\bar{R}^2$ of the model, that is, the adjusted ratio $\frac{\text{variation of } \hat{\mathbf{Y}}}{\text{variation of } \mathbf{Y}}$, expresses the proportion of total community variation that is accounted for under the constraint of the stressors, in other words the magnitude of contamination impact on the community. The exact computation of $\bar{R}^2$ from the regression fit is given in Appendix 1.5.2.

In this test, we randomly draw B site sets, each of size m_{ref} , denoted $\mathbf{S}_b$, $b \in \{1, \dots, B\}$. The above multivariate regression model is fitted on each set, yielding a distribution of B values $\bar{R}_b^2$ that reflect the magnitude of contamination impact

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on the community under random sampling. The set of least-contaminated sites comprising the lowest 20th percentile is denoted $\mathbf{S}_{\text{ref}}$; fitting the same model on this set yields a single value $\bar{R}_{\text{ref}}^2$, reflecting the magnitude of contamination impact on the community within the least-contaminated set.

We then compare $\bar{R}_{\text{ref}}^2$ against the B values $\bar{R}_b^2$ obtained by random resampling. If $\bar{R}_{\text{ref}}^2$ is significantly lower than the randomly distributed $\bar{R}_b^2$ values, the level of contamination impact on the community within the least-contaminated set is taken to be significantly lower than that observed under random sampling, thereby validating that contamination stress exerts a lower level of influence on community composition within the least-contaminated set.

The specific computational procedure is summarised in Algorithm 2.

Algorithm 2: Between-group resampling test for reduced contamination impact in least-contaminated sites

Data: Latent-stressor matrix $\mathbf{X}$, community matrix $\mathbf{Y}$, reference size m_{ref} , replicates B .

Result: Significance of reduced contamination impact in $\mathbf{S}_{\text{ref}}$.

- 1 Fit the reference set: take $\mathbf{S}_{\text{ref}}$ as the lowest 20% of sites by SumRel, regress $\mathbf{Y}$ on $\mathbf{X}$, and record $\bar{R}_{\text{ref}}^2$
 - 2 **for** $b \leftarrow 1$ **to** B **do**
 - 3 Draw a random site set $\mathbf{S}_b$ of size m_{ref} , fit the same regression, and record $\bar{R}_b^2$
 - 4 Compare $\bar{R}_{\text{ref}}^2$ with the null distribution $\{\bar{R}_b^2\}_{b=1}^B$
 - 5 **return** $\mathbf{S}_{\text{ref}}$ is validated if $\bar{R}_{\text{ref}}^2$ lies significantly below $\{\bar{R}_b^2\}$
-

This comparison constitutes a nonparametric, resampling-based hypothesis test. Rather than referring $\bar{R}_{\text{ref}}^2$ to a theoretical sampling distribution, we construct an empirical null distribution $\{\bar{R}_b^2\}_{b=1}^B$ directly by random resampling, so that no parametric assumptions on the distribution of the test statistic are required. The hypotheses are stated in Table 1.1. Under H_0 , the least-contaminated set behaves no differently from an arbitrary set of the same size, and $\bar{R}_{\text{ref}}^2$ is an ordinary draw from the null distribution; under the one-sided alternative H_A , the least-contaminated set exhibits weaker stressor-constrained variation, so that $\bar{R}_{\text{ref}}^2$ falls in the lower tail of $\{\bar{R}_b^2\}$. The

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corresponding left-tailed empirical p -value is

$$p = \frac{1 + \#\{b : \bar{R}_b^2 \leq \bar{R}_{\text{ref}}^2\}}{B + 1},$$

and H_0 is rejected at level α when $p \leq \alpha$, validating that contamination stress exerts a significantly weaker influence on community composition within the least-contaminated set than under random sampling.

Table 1.1: Hypotheses for the resampling-based test of reduced stressor-constrained variation in the least-contaminated set.

H_i	Statement
H_0	The m_{ref} lowest-scored sites do not show weaker stressor-constrained variation than a random m_{ref} -site subset.
H_A	The m_{ref} lowest-scored sites show weaker stressor-constrained variation than a random m_{ref} -site subset.

1.2.5 Within-group permutation test

The between-group test above asks whether the least-contaminated set stands out *relative to other sets of equal size*. A complementary question can be posed about a single set in isolation: judged only against its own data, does the set carry a real stressor–community association, or none at all? We address this with a within-group permutation test.

For an observed set $\mathbf{S}_b$, we randomly permute the rows of its community matrix $\mathbf{Y}_b$ a total of B' times. Each permutation breaks the site-wise correspondence between stressors and community records, so the refitted value $\bar{R}_b^{2,\text{perm}}$ measures the stressor-constrained variation attainable by chance alone. The B' permutations form a null distribution against which the observed $\bar{R}_b^2$ is judged.

For the least-contaminated set, we are interested in the opposite outcome to a conventional significance test: the set qualifies as a natural benchmark precisely when its stressor–community association cannot be distinguished from chance. If the observed $\bar{R}_b^2$ does *not* lie in the upper tail of the permutation distribution—that is,

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if it is no more extreme than the constrained variation produced by random row reassignment—then the contamination gradient exerts no detectable constraint on community composition within the set. The corresponding right-tailed permutation p -value,

$$p_b = \frac{1 + \#\{j : \bar{R}_{b,j}^{2,\text{perm}} \geq \bar{R}_b^2\}}{B' + 1},$$

is the probability of obtaining an association at least as strong as the observed one when stressors and community are in truth unrelated. A *large* p_b therefore indicates that the observed association is indistinguishable from random reassignment, so that we can **not** reject the null hypothesis: the set carries essentially no contamination-constrained structure. Ecologically, failing to reject the null hypothesis is desirable: it characterises a community organised independently of the contamination gradient—whereas a small p_b would instead reveal a set within which contamination shapes community composition. The hypotheses tested here can be expressed in the form of Table 1.2, and the detailed computation can be summarised in the form of Algorithm 3.

Table 1.2: Hypotheses for the within-group permutation test of stressor-constrained variation. A least-contaminated set is expected to *fail to reject* H_0 , indicating no detectable contamination constraint on community composition.

H_i	Statement
H_0	Within $\mathbf{S}_b$, the stressors constrain community variation no more strongly than under random row reassignment of $\mathbf{Y}_b$.
H_A	Within $\mathbf{S}_b$, the stressors constrain community variation significantly more strongly than under random row reassignment of $\mathbf{Y}_b$.

Algorithm 3: Within-group permutation test of stressor–community association

Data: Site set $\mathbf{S}_b = (\mathbf{X}, \mathbf{Y}_b)$, permutations B' .

Result: Significance of the stressor–community association within $\mathbf{S}_b$.

- 1 Fit the observed set: regress $\mathbf{Y}_b$ on $\mathbf{X}$ and record $\bar{R}_b^2$
 - 2 **for** $j \leftarrow 1$ **to** B' **do**
 - 3 Permute the rows of $\mathbf{Y}_b$ to obtain $\mathbf{Y}_b^{\text{perm}}$, refit on $\mathbf{X}$, and record $\bar{R}_{b,j}^{2,\text{perm}}$
 - 4 Compute the right-tailed p -value p_b from $\bar{R}_b^2$ and $\{\bar{R}_{b,j}^{2,\text{perm}}\}_{j=1}^{B'}$
 - 5 **return** $\mathbf{S}_b$ behaves as a natural benchmark if $p_b > \alpha$ (the contamination constraint is indistinguishable from random reassignment)
-

Together, the between-group resampling test of Algorithm 2 and the within-group permutation test of Algorithm 3 probe the least-contaminated set from two complementary angles—its standing relative to comparable sets, and the strength of its own internal stressor–community signal. By scanning candidate sets of increasing size ordered by SumRel and tracking the size at which p_b falls below α , we can observe how the size of least-contaminated group affects the testing outcomes and have insights for choosing a defensible group size.

The scanning procedure is an extension of Algorithm 2 and 3, increasing the least-contaminated group size from 15 to 310 sites in stepwise increments of 5 sites and performing the two tests at each step. At every size m_{ref} , the reference curves $\bar{R}_{\text{ref}}^2$ and p_{ref} are recorded together with the random-subset bands of the same statistics, as summarised in Algorithm 4.

After we apply this procedure across cases, it will show an informative result: in the *corridor case*, where lightly contaminated sites are comparatively abundant, it supports a relatively large least-contaminated set, so that the 20% threshold is a reasonable choice; in the *Detroit River case*, where such sites are scarcer, it favours a smaller set, so that the same 20% threshold becomes difficult to justify.

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Algorithm 4: Threshold scan across least-contaminated subset sizes

Data: Latent-stressor matrix $\mathbf{X}$, community matrix $\mathbf{Y}$, SumRel-ordered
sites, step $\delta = 5$.

Result: Curves $\bar{R}_{\text{ref}}^2(m_{\text{ref}})$ and $p_{\text{ref}}(m_{\text{ref}})$ with random-subset bands.

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1 for  $k \leftarrow 0$  to 59 do
2    $m_{\text{ref}} \leftarrow 15 + k \delta$ 
3   Take  $\mathbf{S}_{\text{ref}}$  as the  $m_{\text{ref}}$  lowest-SumRel sites, and record  $\bar{R}_{\text{ref}}^2$  and  $p_{\text{ref}}$  with
   their random-subset bands (Algorithms 2–3)
4 return  $\{\bar{R}_{\text{ref}}^2(m_{\text{ref}})\}$  and  $\{p_{\text{ref}}(m_{\text{ref}})\}$ 

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1.3 Results

1.4 Tables and Figures

Table 1.3: Rotated PCA loadings and variance summary for the first six principal components

Chemicals	PC1	PC2	PC3	PC4	PC5	PC6
Cr	0.9174	0.2387	0.1500	-0.0043	-0.0663	0.0623
Zn	0.8865	-0.0024	-0.0226	-0.0688	0.1449	0.0614
Ni	0.8861	0.0193	-0.0291	0.0311	-0.2636	-0.0020
Cu	0.8839	0.2188	0.1136	-0.0860	-0.0045	-0.0379
total PCB	0.8364	-0.3031	0.0049	0.1665	0.1523	0.0210
Cd	0.7387	-0.0877	0.4223	0.0595	0.1766	0.1071
Co	0.6772	0.6583	0.0279	0.0688	0.1882	0.0637
p,p'-DDE	0.6385	-0.1622	-0.1223	0.4012	0.2385	0.0789
Mn	0.5012	0.3860	0.3665	0.1539	-0.3701	0.1585
Ca	0.2516	0.1462	0.0287	0.1107	0.0568	0.0954
Al	0.4609	0.8085	0.0437	0.0214	0.1118	0.1404
Pb	0.3790	-0.0207	0.8037	0.0162	-0.3245	-0.0084
As	0.3145	0.1097	0.7899	-0.0134	0.3535	0.0510
OCS	0.1816	0.0541	0.0238	0.9356	0.0096	0.0095
Hg	0.4581	0.2655	0.0210	0.0609	0.7510	0.0320
Fe	0.4722	0.1712	0.0579	0.0212	0.0190	0.8420
Explained Var.	6.5407	1.6036	1.6422	1.1264	1.1890	0.7980
Prop. Var.	0.4088	0.1002	0.1026	0.0704	0.0743	0.0499
Cum. Prop.	0.4088	0.5090	0.6117	0.6821	0.7564	0.8063

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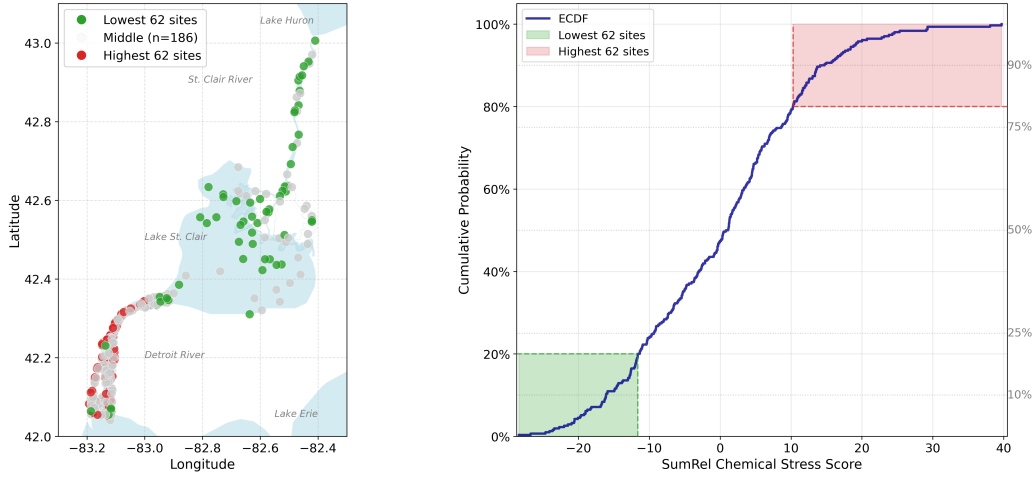


Figure 1.1: Sum Relative Contamination Score (SumRel) across the 310 corridor sites. Left: corridor map of the lowest 20% (62 sites, green), the highest 62 sites (red), and the middle 186 sites (grey). Right: SumRel ECDF, with the lowest and highest 62 sites bifurcated at the 20th and 80th percentiles.

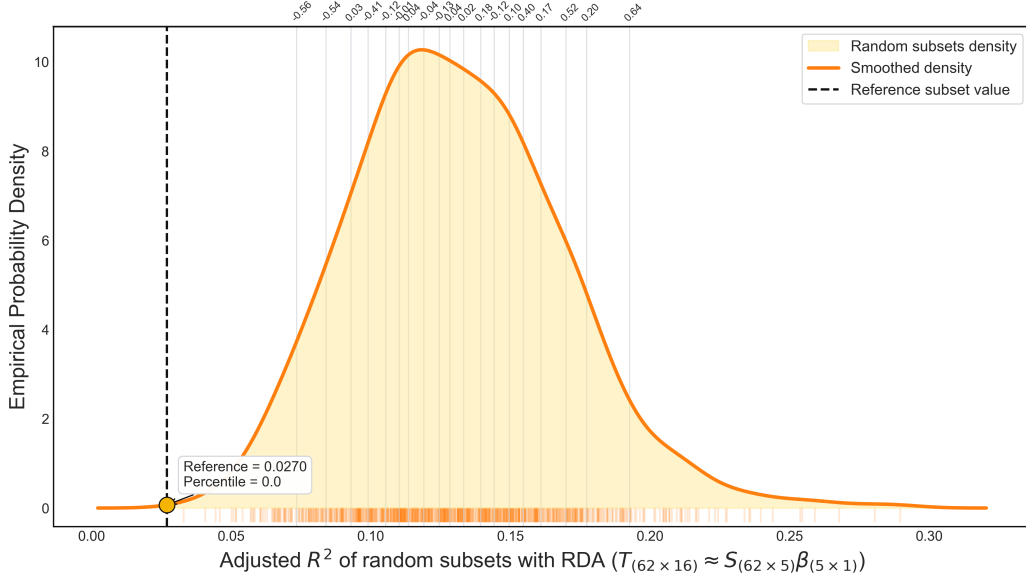


Figure 1.2: Between-group resampling test (Algorithm 2) at $m_{\text{ref}} = 62$: empirical $\bar{R}^2$ distribution of 1000 random 62-site subsets. The top axis shows average contamination scores of the 50 nearest subsets per location, with tick density proportional to probability mass. The black line marks the reference $\bar{R}^2$ (percentile = 0.0); with $p \leq 0.05$, H_0 is rejected—the least-contaminated set is significantly cleaner than random sets.

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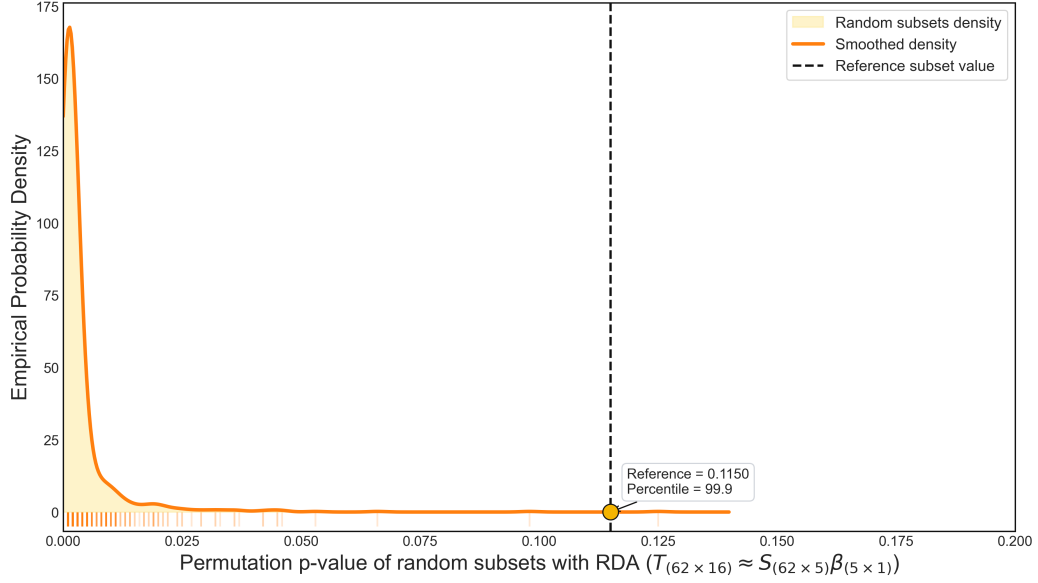


Figure 1.3: Within-group permutation test (Algorithm 3) at $m_{\text{ref}} = 62$: empirical distribution of permutation p -values from 1000 resampled subsets, each permuted 1000 times. The black line marks the reference p -value (percentile = 99.9); with $p \geq 0.05$, H_0 is not rejected—within the least-contaminated set the constraint is indistinguishable from chance, as expected of a natural benchmark.

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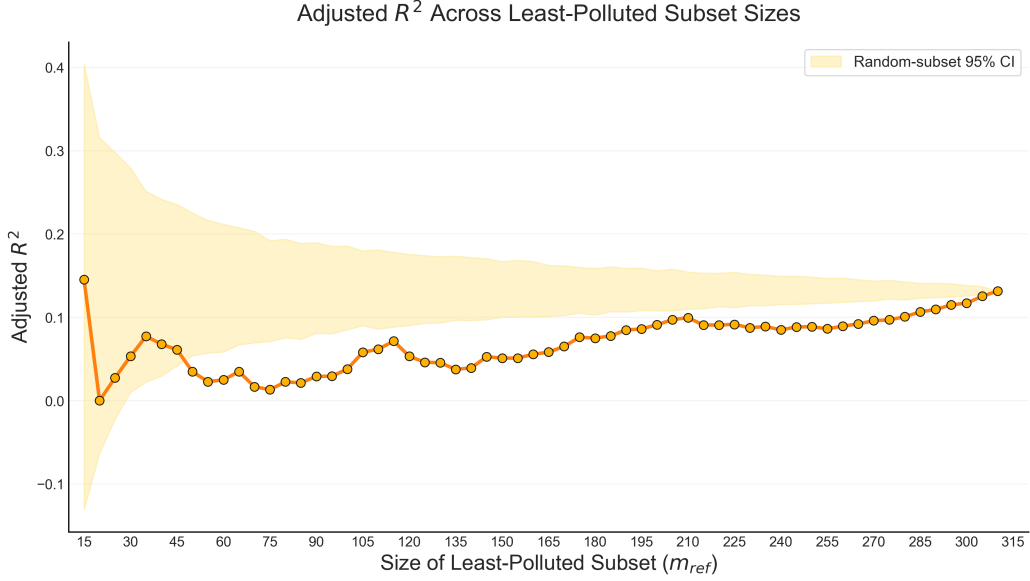


Figure 1.4: Threshold scan of $\bar{R}^2$ across subset sizes (Algorithm 4). The orange curve traces $\bar{R}_{ref}^2$ of the m_{ref} lowest-SumRel sites for $m_{ref} = 15$ to 310; the band is the random-subset 95% interval. The curve hugs the lower edge at small sizes and rises toward the band centre as more contaminated sites are admitted.

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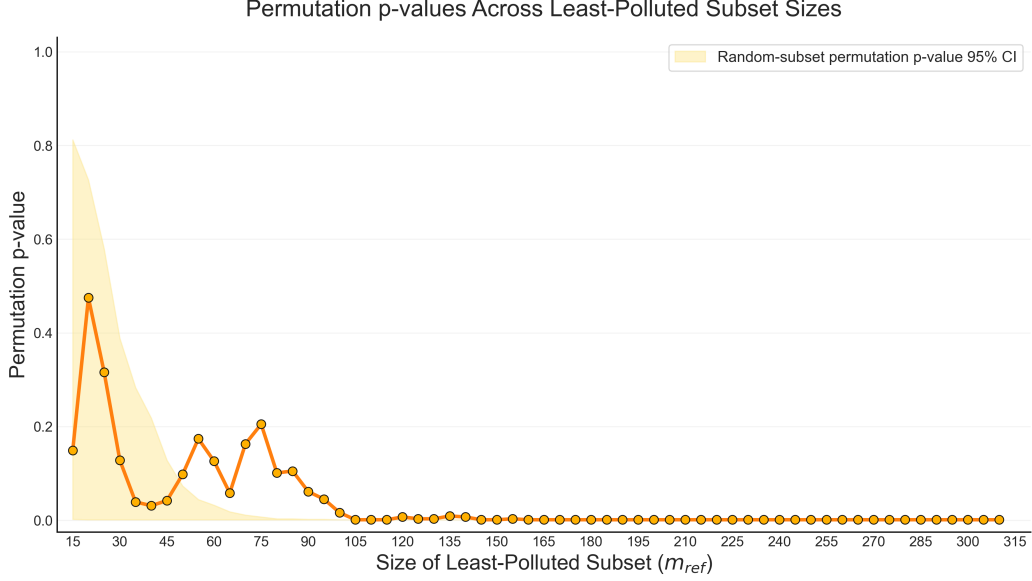


Figure 1.5: Threshold scan of the permutation p -value across subset sizes (Algorithm 4). The orange curve traces p_{ref} of the m_{ref} lowest-SumRel sites for $m_{\text{ref}} = 15$ to 310; the band is the random-subset 95% interval. The reference p_{ref} stays high for small sets and collapses toward zero beyond $m_{\text{ref}} \approx 100$, marking where contamination begins to constrain community composition.

1.5 Appendix

1.5.1 Orthogonality and variance preservation of principal components

Setup. Let $\mathbf{X}$ be an $n \times p$ data matrix recording p descriptors (variables) over n objects (sites). Assume each column has been centred and standardised to unit variance, so that the $p \times p$ correlation matrix of the column space is

$$\mathbf{R} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}.$$

By construction $\mathbf{R}$ is *symmetric* ($\mathbf{R}^\top = \mathbf{R}$) and *positive semi-definite*, since for any vector $\mathbf{u} \in \mathbb{R}^p$,

$$\mathbf{u}^\top \mathbf{R} \mathbf{u} = \frac{1}{n-1} (\mathbf{X}\mathbf{u})^\top (\mathbf{X}\mathbf{u}) = \frac{1}{n-1} \|\mathbf{X}\mathbf{u}\|^2 \geq 0.$$

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Step 1 — Eigen-decomposition exists and yields an orthonormal basis. By the spectral theorem, every real symmetric matrix admits a decomposition

$$\mathbf{R} \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad k = 1, \dots, p,$$

with *real* eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ (non-negative by positive semi-definiteness). The eigenvectors are mutually orthogonal: for $\lambda_i \neq \lambda_j$,

$$\lambda_i \mathbf{v}_i^\top \mathbf{v}_j = (\mathbf{R} \mathbf{v}_i)^\top \mathbf{v}_j = \mathbf{v}_i^\top \mathbf{R} \mathbf{v}_j = \lambda_j \mathbf{v}_i^\top \mathbf{v}_j \implies (\lambda_i - \lambda_j) \mathbf{v}_i^\top \mathbf{v}_j = 0 \implies \mathbf{v}_i^\top \mathbf{v}_j = 0.$$

Hence we may normalise them into an orthonormal set, collected as columns of an orthogonal matrix $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_p]$ with $\mathbf{V}^\top \mathbf{V} = \mathbf{V} \mathbf{V}^\top = \mathbf{I}$, giving

$$\mathbf{R} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p).$$

Step 2 — The eigenvectors preserve the structure of $\mathbf{R}$. Because $\mathbf{V}$ is orthogonal, the decomposition is an *exact* change of basis: no information in $\mathbf{R}$ is lost, only re-expressed. The eigenvectors form the directions along which $\mathbf{R}$ acts purely by scaling, and the total variance is conserved,

$$\text{tr}(\mathbf{R}) = \text{tr}(\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top) = \text{tr}(\mathbf{\Lambda}) = \sum_{k=1}^p \lambda_k = p,$$

so each λ_k is precisely the amount of total variance carried by direction $\mathbf{v}_k$. The structure of $\mathbf{R}$ is thus repackaged, without distortion, into ordered orthogonal axes.

Step 3 — Projection gives orthogonal scores that retain maximal variation. Project the descriptors onto the eigenvectors to obtain the principal-component scores

$$\mathbf{F} = \mathbf{X} \mathbf{V}, \quad \mathbf{f}_k = \mathbf{X} \mathbf{v}_k.$$

These new axes are *mutually uncorrelated*: their covariance matrix is diagonal,

$$\frac{1}{n-1} \mathbf{F}^\top \mathbf{F} = \mathbf{V}^\top \left(\frac{1}{n-1} \mathbf{X}^\top \mathbf{X} \right) \mathbf{V} = \mathbf{V}^\top \mathbf{R} \mathbf{V} = \mathbf{\Lambda},$$

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so $\text{Var}(\mathbf{f}_k) = \lambda_k$ and $\text{Cov}(\mathbf{f}_i, \mathbf{f}_j) = 0$ for $i \neq j$. Because the λ_k are ordered, the first few components carry the largest share of variation: retaining the leading $m < p$ axes captures the fraction

$$\frac{\sum_{k=1}^m \lambda_k}{\sum_{k=1}^p \lambda_k}$$

of the total variance, the maximum attainable by any m orthogonal directions (Rayleigh–Ritz). The optionally *scaled* eigenvectors $\mathbf{v}_k \sqrt{\lambda_k}$ (distance/correlation biplot scaling) rescale axis length to reflect λ_k while leaving orthogonality untouched.

1.5.2 Computation of the adjusted constrained variation $\bar{R}^2$

Setup. Consider a site set of m objects with latent-stressor matrix $\mathbf{X} \in \mathbb{R}^{m \times p}$ (an intercept column prepended) and Hellinger-transformed, column-centred community matrix $\mathbf{Y} \in \mathbb{R}^{m \times q}$ over q taxa. The aim is to measure how much community variation the stressors constrain.

Step 1 — Multivariate least-squares fit. Regressing all taxon columns of $\mathbf{Y}$ on $\mathbf{X}$ gives the fitted values

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top,$$

where $\mathbf{H}$ projects onto the column space of $\mathbf{X}$; $\hat{\mathbf{Y}}$ is thus the portion of the community response lying within the stressor space.

Step 2 — Variation partitioning. With $\mathbf{H}$ symmetric and idempotent, the explained fraction of community variation is

$$R^2 = \frac{\text{SS}_{\text{fit}}}{\text{SS}_{\text{tot}}} = \frac{\text{tr}(\mathbf{Y}^\top \mathbf{H} \mathbf{Y})}{\text{tr}(\mathbf{Y}^\top \mathbf{Y})}.$$

Step 3 — Adjustment for dimensionality. Since R^2 rises with p and falls with m , sets of differing size are not directly comparable; Ezekiel’s degrees-of-freedom correction (Ezekiel, 1941 [6]) gives

$$\bar{R}^2 = 1 - (1 - R^2) \frac{m - 1}{m - p - 1}.$$

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This $\bar{R}^2$ is the statistic compared across reference, random, and permuted sets of common size m_{ref} , so that differences reflect the strength of the stressor–community constraint rather than the dimensions of the fit.

Chapter 2

Natural Community Types: Partitioning Natural Variation and Associating with Environmental Descriptors

2.1 Introduction

Before the contamination stress influence on the community can be examined in isolation, the influence of the natural environment must be taken into account. The community records of interest at any point in space and time are the joint outcome of natural environmental conditions, contamination pressure, and other, unrecorded sources of stress(Allan, 2004 [1]). Therefore a site's taxa record is a representation of all these factors acting together.

Two strategies are commonly used: adjusting for the natural environment within the model, or stratifying the data by it before the model is fitted. The first introduces the environmental variables as covariates in the regression of the community records of interest, so that contamination and natural conditions are modelled jointly and estimated finely; its drawback is that contamination gradients typically covary with natural environmental gradients (Graham, 2003 [7]), resulting collinearity with un-

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stable coefficients and limited practical value. The second strategy instead treats the natural environment as a categorical descriptor and partitions the sites into classes of similar environmental and(or) regional—characteristics (Hughes, Larsen, and Omernik, 1986 [10]; Allan, 2004 [1]) to avoid introducing covariates. This stratification can be a more interpretable approach, since it isolates the contamination signal within definable and comparable habitat stratas and divides the community records of interest with natural variation into community types with aligned environmental conditions.

However, the difficulty is practical rather than conceptual. Natural community composition is shaped by a large set of environmental features, many of which are unobservable, observable but not recorded, or recorded only with substantial measurement uncertainty. Consequently, in most empirical settings, the limited set of measured environmental variables is insufficient to define site groups that are genuinely homogeneous in both environmental conditions and the community records of interest. This limitation is illustrated in Figure 2.1. Complete environmental observation aligns the partition in Ω_{env} with the true community boundary; incomplete observation yields only a degraded partition that mislabels the sites between it and the true boundary—an error of incomplete descriptors, not of a failed environment–community relationship.

For this reason, we reverse the order in which environment and community enter the stratification. We first let the community data *define* the types directly, and then ask what environmental signature *describes* them. It rests on the previous conclusion: within the least-contaminated set, the recorded assemblages retain natural community composition. Therefore we can read natural community types from these least-contaminated sites, since they carry natural variation that we wish to stratify by. Starting with the least-contaminated set, we cluster its community matrix, validate and retain K branches as natural community types. Next, we model this environment-assemblage association by training a classifier that maps their environmental descriptors onto the defined community types. Through this process, we uncanonically stratify community types on the taxa space Ω_{taxa} , makes the implicit environment-determination assumption become an explicit and measurable property in the classifier building. This *define-then-describe* workflow is illustrated in Fig-

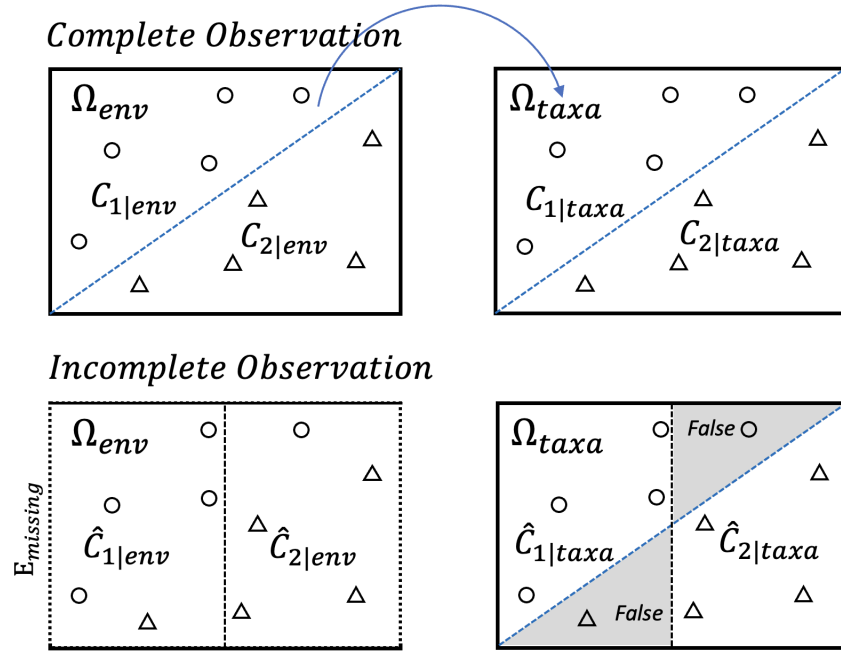


Figure 2.1: Under the assumption that environment determines community, incomplete environmental observation propagates classification error from Ω_{env} to Ω_{taxa} . Complete observation recovers the true diagonal boundary; incomplete observation imposes a degraded vertical partition, mislabelling sites in the wedge between the two (shaded, “False”).

ure 2.2.

The built classifier is then applied to the remaining, contaminated sites: predicting their environmental descriptors into community types. Through this process, every site regardless of contamination status is assigned to its natural community type in a broad sense.

2.2 Methods

The overall aim of this stage is to assign every site, regardless of contamination level, to a natural community type. The procedure has three main steps, the first two carried out entirely within the least-contaminated set and the third applying their results to the remaining sites. First, natural community types are defined on the least-contaminated sites by clustering their community matrix. Second, starting from these

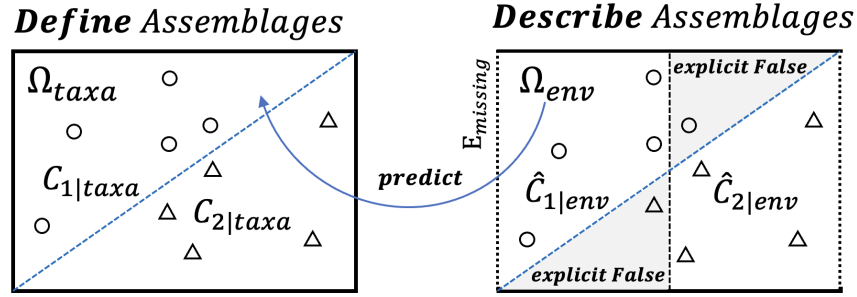


Figure 2.2: Community types are *defined* on the taxa space Ω_{taxa} , then *described* and predicted from the environmental space Ω_{env} . Limited environmental features make some misassignments, but these are rendered explicit (shaded, “explicit False”).

defined types, a discriminating subset of environmental descriptors is identified and a classifier is built to quantify the relationship between environmental features and community types. Third, the classifier is applied to the remaining contaminated sites, predicting and delineating the decision boundary by which they are assigned to the corresponding natural community type. The remainder of this section describes each step in detail.

2.2.1 Unconstrained community typing and its statistical validation

Dissimilarity between sites, Double-zero problem and Hellinger transformation

Hierarchical clustering operates on a matrix of pairwise dissimilarities, so the first choice is how dissimilarity between sites is measured from the community records. Applying the Euclidean distance directly to raw abundances is inappropriate for community data, because it treats the joint absence of a taxon at two sites as evidence of similarity—the *double-zero problem*—and lets a few abundant taxa dominate the distance. To avoid this, the community matrix is first Hellinger-transformed: at a given site, each taxon’s record is divided by the site’s total and the square root is taken, so that the entry becomes the square root of the taxon’s relative abundance at that site. The Euclidean distance computed between two Hellinger-transformed site

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profiles is then equal to the Hellinger distance between their original compositions, a measure that is insensitive to double zeros and to differences in total abundance, and which is recommended for ordination and clustering of community data in numerical ecology (Borcard, Gillet, and Legendre, 2018 [4]). This transform therefore makes the subsequent Euclidean-based clustering ecologically meaningful.

Hierarchical clustering with Ward’s method

The Hellinger-based dissimilarities are clustered by agglomerative hierarchical clustering. Among the available linkage functions, Ward’s method is chosen because it merges, at each step, the pair of clusters whose fusion produces the smallest increase in within-cluster variance of the Euclidean (here Hellinger) distances. This criterion suits the goal of delineating community types with members mutually similar in composition. The procedure yields a dendrogram in which the height of each fusion records the linkage function value across which the corresponding branches join; cutting the dendrogram at a chosen linkage height then defines the K community types retained for the subsequent analysis.

Statistical evidence for the branching: node-wise and overall ANOVA

A dendrogram will partition any data, so the branching structure requires statistical support. Two complementary tests are used, both applied to the Hellinger-transformed taxa abundances used in the clustering rather than to the raw counts. First, a *node-wise* ANOVA is applied from the first to $K + 2$ branching nodes of the dendrogram: for every taxon, its Hellinger-transformed abundances in the two daughter branches are compared, identifying the taxa whose mean values differ significantly across the split. These taxa are the indicator species that contribute most to the formation of that particular node and drive the bifurcation. Second, once the K community types are finalised, an overall ANOVA tests, for each taxon, whether its Hellinger-transformed mean abundance differs significantly among the K groups, providing statistical evidence that the retained partition separates the community types along interpretable compositional gradients.

Robustness of the partition with pvclust

Beyond the compositional evidence for each branch, the stability of the branches under resampling is assessed with pvclust (Suzuki and Shimodaira, 2006 [16]; Suzuki, Terada, and Shimodaira, 2019 [17]). The method repeatedly perturbs the descriptor space by multiscale bootstrap resampling of the taxa, re-runs the same clustering procedure on each resampled space, and checks whether each branch of the original (anchor) dendrogram reappears. From the frequency with which a branch recurs across the B bootstrap replicates, an *Approximately Unbiased* (AU) p -value is computed for that branch. A high AU p -value means the branch is recovered in a large proportion of the perturbed spaces—its existence does not depend on a few particular taxa but is supported by many. The ordinary *Bootstrap Probability* (BP), estimated from fixed-size bootstrap resampling alone, is also reported, but it is a more biased than the AU value. Branches with low AU support are treated as uncertain and inform the choice of where the dendrogram is cut. Figure 2.3 illustrates the computation workflow ¹, more details about this procedure can be found in the articles by Suzuki and Shimodaira (Suzuki and Shimodaira, 2006 [16]).

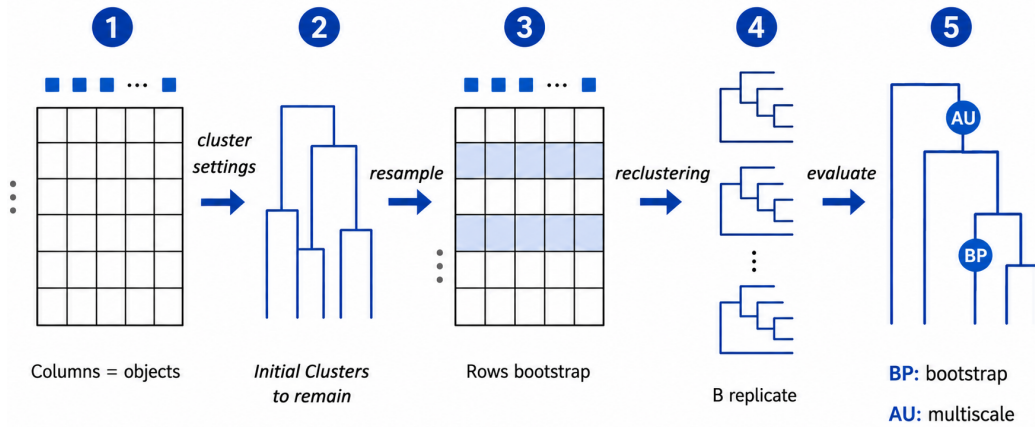


Figure 2.3: The pvclust workflow: bootstrap-resample taxa, recluster B times, and score each anchor branch by its AU (multiscale) and BP (bootstrap) recurrence.

¹The data matrix is transposed so that taxa occupy the rows, because resampling routines in R and similar environments operate row-wise; bootstrapping the feature variables therefore requires placing them as rows.

2.2.2 Linking community types to environmental descriptors: classifier training under limited features

Taxa clusters entangled in environmental space, and the 2008 solution

A key difficulty of the environmental stratifying strategy persists in this design: the limited set of measurable environmental features. Types delineated by unconstrained clustering may remain entangled in environmental space, where one or more missing key variables, or attention to uninformative ones, blurs the environment–taxa cluster relationship. This already surfaced in the 2008 corridor study (Zhang, 2008 [18]).

There, the Ward-clustered community types could not be cleanly separated on the environmental descriptors, so a directly trained classifier would have performed poorly. The solution was to define the two types by a selective assignment of branches. As shown in Figure 2.8, the dendrogram first divides into B_2 versus the pair $\{B_1, A\}$, and at a lower height do B_1 and A separate. Cluster C_1 was assigned to B_2 and B_1 jointly, and Cluster C_2 to A alone—a hindsight operation under which the types separate more readily on the available descriptors.

The cost, however, is that the types no longer reflect the dendrogram. The assignment is *non-monophyletic*: C_1 joins B_2 and B_1 , two branches that do not form a single sub-tree, while the topologically closer A is placed in C_2 . No single horizontal cut can produce this grouping; the labels are assigned across the tree’s natural splitting order. This solution thus lets environmental separability override the community structure the clustering reveals (essentially induced the environmental-stratifying approach). As recognized an explicit property to reflect the environment-determination assumption, the classifier’s resulting accuracy is inflated and loses its realistic value. (Figure 2.9).

In the environmental descriptor-space view, this operation amounts to relabelling the clustering-defined sites: circles (C_1) that sit, on the available features, among the triangles (C_2) are reassigned to C_2 , buying environmental separability at the cost of the true taxa labels (Figure 2.4 ²).

²The true boundary (blue dashed) is governed by unobserved variables $E_{missing}$ so it is not a diagonal but leans to the $E_{available}$ direction.

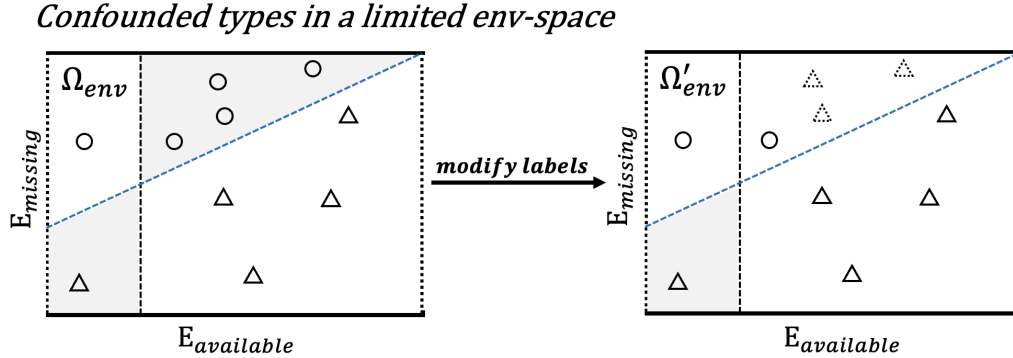


Figure 2.4: Confounded community types in a limited environmental space, illustrating the 2008 approach of Zhang. This solution relabels part of C_1 (circles) as C_2 (triangles) in one direction only ($\Omega_{env} \rightarrow \Omega'_{env}$, dotted triangles), reducing but not removing the overlap—separability bought by altering the true cluster labels.

Describing fixed taxa types: the sibling-separation criterion

The 2008 solution failed by adjusting the labels to the environment. The difficulty it faced, however, is real and remains: the community types defined in the taxa space Ω_{taxa} do not necessarily appear as distinct classes in Ω_{env} . Rather than adding descriptors or relabelling, this framework keeps the two stages separate: the community types are defined first from taxa composition, and then described by environmental features, under the principle that the description stage must not influence the definition stage. We treat the taxa-defined clustering as primary, because it is tied directly to the ecological target, and ask only which environmental descriptors best reflect that fixed typology. The objective at this stage is not to explain the variation of individual taxa, but to separate a small number of assemblage types—coarser objects than the original taxa matrix, and therefore potentially distinguishable with a smaller, more targeted subset of descriptors.

This rests on a general principle: *the finer the typology, the more environmental information is usually required to separate its classes*, and not all descriptors help—some only add noise. Consider grouping vehicles into two types, those moving above the surface (aircraft) and those moving on it (cars). Longitude and latitude are then misleading, since an aircraft over a city and a car near an airport share horizontal coordinates, whereas the same aircraft over the ocean is well separated by them;

the discriminating power of a descriptor depends on how the cases are sampled. Dropping longitude and latitude and keeping the single key descriptor, elevation, separates the two types cleanly (Figure 2.5). Introducing a third type—ships, also on the surface but distinct from cars—would make elevation alone insufficient and bring the horizontal descriptors back into use: a finer typology demands a richer subset.

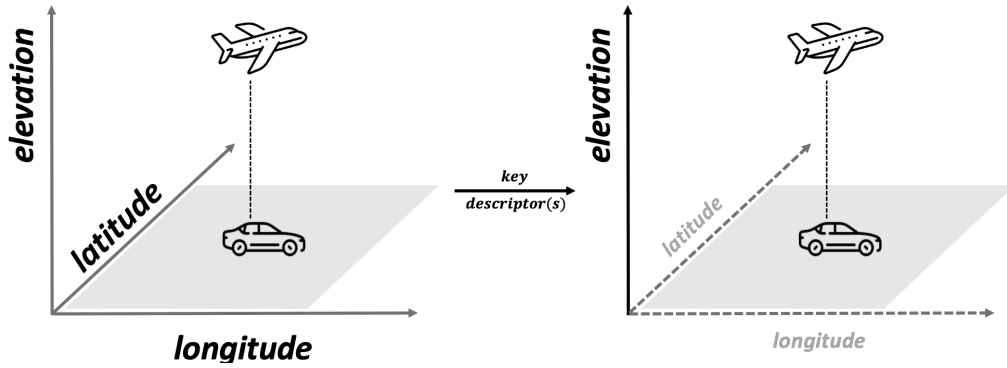


Figure 2.5: Not all descriptors help separate classes. An aircraft and a car are barely distinguishable on the horizontal descriptors (longitude, latitude), yet a single key descriptor—elevation—separates them cleanly. When a few classes are defined and must be told apart, the right descriptor subset matters more than the number of descriptors, and retaining irrelevant ones only adds noise.

To make this precise, we fix terminology. The retained branches are the *taxa-defined community types*; for a site in type C_k , the other sites in C_k are its *siblings*, and sites in any other type C_j ($j \neq k$) are its *non-siblings*. A descriptor subset spans a space $\Omega_{E'}$, and its quality is judged by a single criterion.

Sibling-separation criterion. For a site in type C_k , its environmental position should lie closer to its siblings in C_k than to its non-siblings in any rival type C_j ($j \neq k$). The more sites that remain closer to their siblings than to any rival type, the better $\Omega_{E'}$ separates the taxa-defined community types.

The logic is direct: if a reliable environment–community relationship holds, sites of one taxa-defined type should form a tight, coherent group in a suitable descriptor space. Group-level coherence is thus an aggregate of how consistently individual sites

stay near their siblings, and the discriminating space $\Omega_{E'}^*$ is the candidate subset that maximises this coherence across all K types (Figure 2.6).

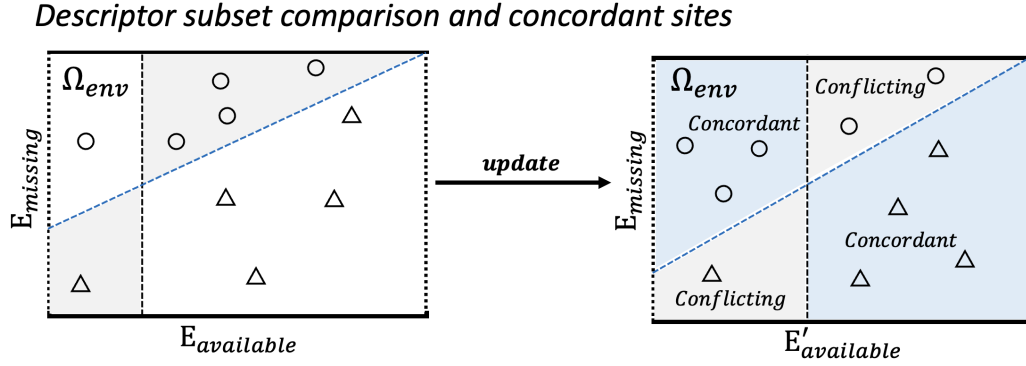


Figure 2.6: Selecting a descriptor subset and identifying concordant sites. Under the initial descriptors $E_{available}$, the taxa-defined types overlap in environmental space (left). A more discriminating subset $E'_{available}$ separates most sites into *concordant* regions (blue), where sites lie close to their siblings, leaving *conflicting* regions (grey) where sites lie closer to non-siblings. Concordant sites represents a majority environment–community relationship; conflicting sites hold the opposite, or expose the limits of, or noise in, the available descriptors.

Single clustering: a preservation measure for exogenous clusters

Following the sibling-separation criterion, we need to quantify, in a candidate space $\Omega_{E'}$, how closely each site stays with its siblings. The building block is a single clustering of the reference sites, from which a site-level concordance margin is derived; the resampling-based extension is deferred to the next section.

From dissimilarity to a single clustering. Closeness between two sites i and i' in $\Omega_{E'}$ is measured by a dissimilarity $d_{E'}(i, i') \geq 0$ —the smaller, the closer. Collecting all $\binom{m_{\text{ref}}}{2}$ pairs gives a dissimilarity matrix $D_{m_{\text{ref}} \times m_{\text{ref}}}^{E'}$, which a clustering algorithm cuts into K groups (the same K as the taxa typology). This single clustering alone is unsupervised: it says how the sites group in $\Omega_{E'}$, but nothing yet about whether they group *with their siblings*. The sibling structure enters only through the external anchor labels C_k defined in Ω_{taxa} .

Co-assignment and sibling counts. The clustering outcome is recorded in a co-assignment matrix $M^{E'}$, with

$$M_{ii'}^{E'} = \begin{cases} 1, & i \text{ and } i' \text{ fall in the same environmental cluster,} \\ 0, & \text{otherwise.} \end{cases}$$

For a site $i \in C_k$, summing $M_{ii'}^{E'}$ over a chosen reference group counts how many of that group cluster together with i . Summing over its siblings (the rest of C_k) and over a rival type C_j ($j \neq k$) gives, respectively, the sibling count and the rival-type count

$$n_i(C_k) = \sum_{\substack{i' \in C_k \\ i' \neq i}} M_{ii'}^{E'}, \quad n_i(C_j) = \sum_{i' \in C_j} M_{ii'}^{E'} \quad (j \neq k).$$

A larger $n_i(C_k)$ means i stays with more of its siblings; a larger $n_i(C_j)$ means it is drawn toward rival type C_j .

Normalisation and the concordance margin. Because types differ in size, the raw counts are normalised by the number of sites i is compared against— $|C_k| - 1$ for siblings (excluding i itself) and $|C_j|$ for a rival type—giving the average co-clustering of i with each type,

$$\bar{M}_i^{(E)}(C_j) = \begin{cases} \frac{1}{|C_k| - 1} \sum_{\substack{i' \in C_k \\ i' \neq i}} M_{ii'}^{E'}, & j = k \quad (\text{siblings}), \\ \frac{1}{|C_j|} \sum_{i' \in C_j} M_{ii'}^{E'}, & j \neq k \quad (\text{rival type}). \end{cases}$$

The *concordance margin* of site i contrasts its closeness to its own type against its closeness to the most competitive rival type,

$$\mu_i^{(E)} = \bar{M}_i^{(E)}(C_k) - \max_{j \neq k} \bar{M}_i^{(E)}(C_j).$$

A positive margin means i co-clusters more with its siblings than with any rival type—it stays in the environmental cluster holding the majority of its own type; a negative margin means a rival type draws it more strongly.

From site margins to type-level preservation. A type C_k is well preserved in $\Omega_{E'}$ when most of its members have positive margins, i.e. when its majority co-clusters together. This is summarised by the preservation score

$$\rho_k^{(E)} = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{1}\{\mu_i^{(E)} > 0\}, \quad k = 1, \dots, K,$$

the proportion³ of C_k 's members staying with their siblings. A site with positive margin is termed *concordant* and one with negative margin *conflicting*, as in Figure 2.6. High $\rho_k^{(E)}$ across all K types means each taxa-defined type retains its majority as a coherent group in $\Omega_{E'}$ —exactly the group-level coherence demanded by the sibling-separation criterion. The discriminating space is then the candidate maximising this coherence, $\Omega_{E'}^* = \arg \max_{E'} \min_k \rho_k^{(E)}$, which protects the weakest type rather than the average.

The whole single-clustering pipeline, from a candidate descriptor space to type-level preservation, can be summarised as

$$(\Omega_{E'}, d_{E'}(\cdot)) \xrightarrow{\text{cut into } K} M_{m_{\text{ref}} \times m_{\text{ref}}}^{E'} \xrightarrow{\text{aggregate w.r.t. } C_{g(i)}} \mu_i^{(E)} \xrightarrow{\text{aggregate within } C_k} \rho_k^{(E)}.$$

This single-clustering pipeline turns a chosen descriptor space into a per-type measure of how well the taxa-defined typology is recovered: sites are clustered once into K groups, each site's co-assignments are aggregated against its anchor labels $C_{g(i)}$ to give the margin $\mu_i^{(E)}$, and the proportion of positive margins within each type gives the preservation $\rho_k^{(E)}$. The discriminating space keeps every type's majority together, $\Omega_{E'}^* = \arg \max_{E'} \min_k \rho_k^{(E)}$.

Consensus clustering: a resampling-based preservation measure

The single clustering of the previous section is fragile: its partition can hinge on a few influential sites, so a site's margin reflects one particular cut rather than a sta-

³Co-assignment in a single clustering is symmetric ($M_{ii'}^{E'} = M_{i'i}^{E'}$), so a positively-margined member of C_k and its siblings are mutually co-clustered, sharing the one environmental group this partition fixes; a high $\rho_k^{(E)}$ thus means the majority of C_k forms one coherent group while the minority disperses.

ble property. Consensus clustering (Monti et al., 2003 [13]) removes this fragility by repeating the clustering over resampled data and reading off how *often* sites group together. The pipeline is unchanged except at its first step: the m_{ref} sites are resampled B times, and at each iteration the dissimilarity $d_{E'}(\cdot)$ is recomputed and the clustering is refitted and cut into K groups.

This single change propagates through the same chain, altering only the co-assignment matrix and, with it, the meaning of every quantity downstream. The single-cut indicator $M_{ii'}^{E'} \in \{0, 1\}$ is replaced by the *consensus* value $\mathbf{M}_{ii'}^{E'} \in [0, 1]$. Writing $b_{ii'}$ for the number of resamples in which both sites i and i' are drawn, and $c_{ii'}$ for the number of those in which they are also assigned to the same cluster, the consensus value is the conditional co-clustering frequency

$$\mathbf{M}_{ii'}^{E'} = \frac{c_{ii'}}{b_{ii'}},$$

taken only over the resamples in which the pair actually co-occurs. A co-assignment is thus no longer a yes/no fact from one partition but a frequency, with a larger value meaning a more stable pairing. The site-level aggregation keeps its form, now reading these frequencies rather than indicators,

$$\bar{\mathbf{M}}_i^{(E)}(C_j) = \begin{cases} \frac{1}{|C_k| - 1} \sum_{\substack{i' \in C_k \\ i' \neq i}} \mathbf{M}_{ii'}^{E'}, & j = k \quad (\text{siblings}), \\ \frac{1}{|C_j|} \sum_{i' \in C_j} \mathbf{M}_{ii'}^{E'}, & j \neq k \quad (\text{rival type}), \end{cases}$$

so that the margin $\mu_i^{(E)} = \bar{\mathbf{M}}_i^{(E)}(C_k) - \max_{j \neq k} \bar{\mathbf{M}}_i^{(E)}(C_j)$ and the preservation score $\rho_k^{(E)} = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{1}\{\mu_i^{(E)} > 0\}$ are defined exactly as before, but now measure stability *under resampling* rather than within one clustering. The robust pipeline is therefore

$$(\Omega_{E'}, d_{E'}(\cdot)) \xrightarrow{B \text{ resamples}} \mathbf{M}_{m_{\text{ref}} \times m_{\text{ref}}}^{E'} \xrightarrow{\text{aggregate w.r.t. } C_{g(i)}} \mu_i^{(E)} \xrightarrow{\text{aggregate within } C_k} \rho_k^{(E)}.$$

Searching the discriminating space. With a resampling-stable $\rho_k^{(E')}$ available for any candidate space, selecting the discriminating subset becomes an optimisation over candidates. Since the per-type scores $\{\rho_k^{(E')}\}_{k=1}^K$ form a vector, we maximise their

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worst-case entry, so that no type is left poorly preserved for the sake of the average,

$$\Omega_{E'}^* = \arg \max_{E'} \min_{1 \leq k \leq K} \rho_k^{(E')}.$$

Searching all 2^p subsets of the p descriptors is straightforward to program but computationally expensive; in practice we evaluated a small number of candidate subsets guided by prior ecological knowledge, choosing the one that preserves the most sites while keeping the preserved proportion balanced across types.

The same check in the taxa view. Because the procedure depends only on the anchor labels $C_{g(i)}$, a feature view Ω_V , and a dissimilarity $d_V(\cdot)$, it applies unchanged to any view $V \in \{T, E\}$, yielding $\mathbf{M}^{(V)}$, $\mu_i^{(V)}$, and $\rho_k^{(V)}$ by the same chain. Applied to the environmental view it gives $\mu_i^{(E)}$, testing whether a site’s taxa-defined label is supported by its environmental position. Applied to the *taxa* view—the very space in which the types were defined—it gives $\mu_i^{(T)}$, a *self-stability* check: it asks whether a site remains with its siblings under resampling of the data that defined its type, so $\mu_i^{(T)} \geq 0$ confirms the site sits stably within its own type. The two views together isolate the informative case: a site with $\mu_i^{(T)} \geq 0$ but $\mu_i^{(E)} < 0$ stays with its siblings in composition yet drifts toward a rival type environmentally—a *conflicting* site, similar in community composition but distinct in environment, and thus a direct expression of insufficient environmental information or ordinary ecological variation. The full cross-view procedure is given in Algorithm 5.

Algorithm 5: Cross-view stability check from a feature view to type-level stability

Data: Feature view Ω_V , dissimilarity $d_V(\cdot)$, anchor labels $C_{g(i)}$, # types K , replicates B .

Result: Consensus matrix $\mathbf{M}^{(V)}$, margins $\{\mu_i^{(V)}\}$, preservation scores $\{\rho_k^{(V)}\}$.

1 **for** $b \leftarrow 1$ **to** B **do**

2 Resample sites with replacement, recompute $d_V(\cdot)$, refit the clustering,
 and cut into K clusters

3 Accumulate co-clustering frequencies into $\mathbf{M}^{(V)} = (\mathbf{M}_{ii'}^{(V)})_{m \times m}$

4 For each site $i \in C_k$, compute $\mu_i^{(V)} = \bar{\mathbf{M}}_i^{(V)}(C_k) - \max_{j \neq k} \bar{\mathbf{M}}_i^{(V)}(C_j)$

5 For each type C_k , compute $\rho_k^{(V)} = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{1}\{\mu_i^{(V)} > 0\}$

6 **return** $\mathbf{M}^{(V)}$, $\{\mu_i^{(V)}\}_{i=1}^m$, $\{\rho_k^{(V)}\}_{k=1}^K$

Consensus clustering II: Learning Majority $E' \rightarrow C_i$ Pattern from Concordant Sites with a Classifier

The cross-view stability check of the previous section assigns each site a pair of margins, $(\mu_i^{(T)}, \mu_i^{(E)})$ that reflects its closeness to majority siblings in two views respectively. To build a classifier that learns a stable environment–community relation from the concordant sites that lie closely to majority siblings in both views, we consider the joint margin signs to group sites into training set or diagnostic set. The four sign combinations and their ecological meaning are summarised in Table 2.1.

Table 2.1: Role of each concordance category in classifier training and diagnostics.

Category	$(\mu^{(T)}, \mu^{(E)})$	Role	Ecological signal
Concordant	$(+, +)$	Training	Baseline mapping
Taxa-only stable	$(+, -)$	Diagnostic	Descriptor insufficiency
Env-only stable	$(-, +)$	Diagnostic	Taxa-label fragility
Ambiguous	$(-, -)$	Diagnostic	Community transition

Concordant sites form the *training set* $\mathcal{S}_{\text{con}} = \{i : (\mu_i^{(T)}, \mu_i^{(E)}) = (+, +)\}$ because

these groups hold most sites that have similar environmental positions under same taxa-defined type, which is the environment–community relation that the classifier worhest to learn. Conflicting sites, in non-concordant groups $\mathcal{S}_{\text{conf}} = \{i : (\mu_i^{(T)}, \mu_i^{(E)}) \neq (+, +)\}$, serve the opposite purpose as *diagnostic set*. These minor conflicting sites have their environmental position closer to a rival type, which is inconsistent with the majority relation observed in the concordant set.

It is important to qualify the term. These sites are not concordant in any absolute sense, but concordant *relative to a given contexts*. They represents a community type and environment relation mutually supported in that context. Being subject to the taxa data, defined community types, and the descriptor space, the concordant set may change accordingly. In the right-hand panel of Figure 2.6, the two blue concordant regions are the majority neighbourhoods of the two types, supporting the association learning from most sites, while the grey conflicting regions hold the minority whose environment matches the other type. If some stated contexts were substantially changed and resulted in most sites moving to the current conflicting region, the concordant and conflicting labels would be reassigned: the region holding the new majority would become concordant, and the minority region conflicting.

We adopt Linear Discriminant Analysis (LDA) as a classical linear classifier to learn and diagnose on the $\mathcal{S}_{\text{con}}$ and $\mathcal{S}_{\text{conf}}$ respectively. Let $\mathbf{E}^*$ denote the environmental feature matrix of sites and $C_{g(i)}$ as their community labels from the original hierarchical clustering. The *concordant classifier* is trained only on the concordant subset $\mathcal{S}_{\text{con}}$:

$$f_{\text{con}} = \text{LDA}\left(\{(\mathbf{E}_i^*, C_{g(i)}) : i \in \mathcal{S}_{\text{con}}\}\right),$$

f_{con} is expected to predict well on held-out concordant sites, since these carry the consistent relation. Applied to the conflicting set $\mathcal{S}_{\text{conf}}$, the f_{con} will perform poorly, or even invert, because those sites contradict the learned pattern.

For comparison, a *benchmark classifier* is trained on all sites regardless of status,

$$f_{\text{bench}} = \text{LDA}\left(\{(\mathbf{E}_i^*, C_{g(i)}) : i \in \mathcal{S}_{\text{con}} \cup \mathcal{S}_{\text{conf}}\}\right).$$

Over all sites regardless of concordance, f_{con} is expected to outperform f_{bench} on the

concordant sites and with a better accuracy in predicting the minority type. But f_{con} underperforms on the conflicting sites as it discarded the contradictory minority information in training. Applying random-split cross-validation to evaluate classifier stability, the f_{con} will have a stable performance in memorizing the pattern it learned from $\mathcal{S}_{\text{con}}$. Whereas the f_{bench} will be poorly stable as it absorbs contradictory information with $\mathcal{S}_{\text{con}} \cup \mathcal{S}_{\text{conf}}$.

The concordant classifier, therefore, represents a majority type–environment relation among the observed sites. It works well at predicting the community type of a new site if (and most likely) it belongs to the majority group. When a new site belongs to the minority conflicting group, it will give misclassification, and this misclassification cannot be detected from the classifier’s own performance. It is revealed only by the consensus-clustering step of the previous section, another relation departs from that of the majority.

Predicting Expected Community Types for Non-reference Sites via Majority Relation

After training and diagnosing f_{con} , we apply it to the remaining non-reference sites to infer the community type, or types, expected from their environmental profiles. For site i with descriptor vector $\tilde{\mathbf{e}}'_i$ in the selected descriptor space $\Omega_{E'}^*$, LDA returns posterior probabilities

$$\pi_{ik} = P(C_k \mid \tilde{\mathbf{e}}'_i), \quad k = 1, \dots, K.$$

Because f_{con} uses only environmental descriptors, two sites with the same environmental profile receive the same posterior distribution. The concordant or conflicting status defined in the previous section is a diagnostic property of least-contaminated sites, obtained from the joint taxa and environmental views. It is not available for other contaminated sites and is not an input to f_{con} . The decision threshold θ therefore does not correct the classifier differently for concordant and conflicting cases. Instead, it expresses the analyst’s tolerance for uncertainty when translating posterior probabilities into community-type assignments.

CHAPTER 2. NATURAL COMMUNITY TYPES: PARTITIONING NATURAL VARIATION AND ASSOCIATING WITH ENVIRONMENTAL DESCRIPTORS

We use θ to form the set of environmentally plausible community types for site i :

$$\hat{\mathcal{C}}_{\theta}(i) = \{ C_k : \pi_{ik} \geq \theta \}.$$

A lower threshold reduces the *risk* of excluding a site's latent or expected community type, but may also admit additional types that are less appropriate. A higher threshold produces a purer assignment, but increases the risk of excluding plausible types. In this work, we give higher priority to avoiding false exclusion, because the subsequent type-specific analyses aim to retain potentially relevant community information rather than to form maximally pure but overly conservative subsets. Accordingly, θ is chosen as a low-to-moderate threshold. The value $\theta = 1/K$ provides an equal-probability baseline and guarantees a non-empty candidate set; in the two-type case, a lower value such as $\theta = 0.3$ allows uncertain sites to retain both plausible labels. Consequently, a site may be assigned to more than one type. The thresholding rule maps the posterior vector

$$\boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{iK})$$

to a binary membership vector:

$$\mathbf{f}_{\theta} : [0, 1]^K \rightarrow \{0, 1\}^K, \quad \mathbf{f}_{\theta}(\boldsymbol{\pi}_i) = \left(\mathbf{1}\{\pi_{ik} \geq \theta\} \right)_{k=1}^K.$$

The support of this vector is exactly the candidate set

$$\hat{\mathcal{C}}_{\theta}(i) = \{ C_k : \pi_{ik} \geq \theta \}.$$

For visualisation, the fitted classifier can be displayed as a decision surface in the selected environmental descriptor space $\Omega_{E'}^*$. To focus on the environmental zone where the observed sites lie, PCA is applied to the sites' E' matrix and returns the first two principal components (the original axes are used directly if $p_{E'} \leq 2$) to form a plane to display decision behavior over these sites. The PCA ordination is used only for visualisation and does not change the fitted classifier. The classifier f_{con} is then evaluated on a grid over the two displayed axes (PC_1 and PC_2), with any remaining components fixed at their mean values. This produces a two-dimensional view of how

environmental profiles are mapped to taxa-defined community types across the region of descriptor space occupied by the sites.

2.3 Results

2.4 Discussion

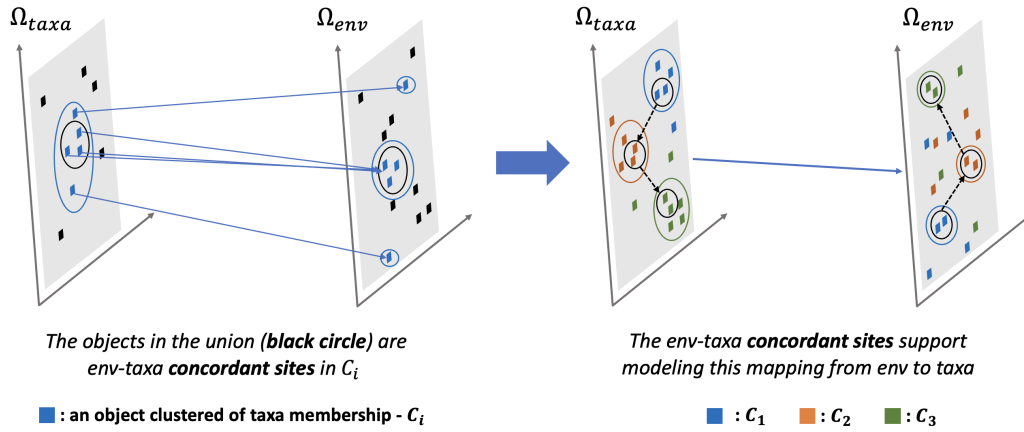


Figure 2.7: to do: this figure is about a more fundamental question of this cross-check request, why did this issue happen and how this cross-check view gives solution to it.

2.5 Figures and Tables

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2.5.1 Appendix

Consensus clustering procedure

Let $\mathbf{X}^{(V)} \in \mathbb{R}^{m_{\text{ref}} \times p_V}$ denote the data matrix of the m_{ref} reference sites in a feature space of view V . In this feature space, a defined dissimilarity measure $d_V(\cdot)$ sets the pairwise dissimilarities between sites and they are stored in $D^{(V)}$. The predefined class labels for all site $i \in \{1, \dots, m\}$ are given by $g(i) \in \{1, 2, \dots, K\}$. The original purpose of consensus clustering is to measure the resampling stability of pairwise (i, i') associations in the chosen view V .

For each bootstrap iteration $b = 1, \dots, B$, a row-resampled version of $\mathbf{X}^{(V)}$ is generated by sampling the m sites with replacement. The resampled matrix follows the same process and protocol as the original $\mathbf{X}^{(V)}$ to compute the pairwise dissimilarities:

$$\mathbf{D}^{(V;b)} = d_V(\mathbf{X}^{(V;b)}),$$

A selected hierarchical clustering rule is applied to $\mathbf{D}^{(V;b)}$, and its resulting dendrogram is cut into K clusters. This produces a bootstrap partition

$$\mathcal{P}^{(V;b)} = \{C_1^{(V;b)}, \dots, C_K^{(V;b)}\}.$$

Across the B bootstrap partitions, the stability of each site pair (i, i') is summarised by its co-assignment frequency. Let

$$I_{ii'}^{(b)} = \mathbf{1}\{i \text{ and } i' \text{ are both sampled at iteration } b\},$$

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and

$$A_{ii'}^{(V;b)} = \mathbf{1}\{i \text{ and } i' \text{ are both sampled and assigned to the same cluster in } \mathcal{P}^{(V;b)}\}.$$

The consensus matrix for view V is then defined as

$$M_{ii'}^{(V)} = \frac{\sum_{b=1}^B A_{ii'}^{(V;b)}}{\sum_{b=1}^B I_{ii'}^{(b)}}, \quad i, i' = 1, \dots, m_{\text{ref}}.$$

Thus,

$$\mathbf{M}^{(V)} = (M_{ii'}^{(V)}) \in [0, 1]^{m_{\text{ref}} \times m_{\text{ref}}}$$

is the output of the consensus clustering procedure. A large value of $M_{ii'}^{(V)}$ indicates that sites i and i' are repeatedly assigned to the same cluster under resampling in view V , whereas a small value indicates an unstable or incidental association.

The resulting consensus matrix $\mathbf{M}^{(V)}$ stores the pairwise stability of sites and supports computing concordance margins for each site by aggregating the pairwise co-assignment values with its siblings and non-siblings that are defined by the anchor partition $g(i)$. For site i with anchor label C_k , its concordance margin $\mu_i^{(V)}$ is computed as:

$$\mu_i^{(V)} = \underbrace{\frac{1}{|C_k| - 1} \sum_{\substack{i' \in C_k \\ i' \neq i}} M_{ii'}^{(V)}}_{\text{mean closeness to siblings}} - \underbrace{\max_{j \neq k} \frac{1}{|C_j|} \sum_{i' \in C_j} M_{ii'}^{(V)}}_{\text{strongest closeness to a rival type}}$$

The higher the margin $\mu_i^{(V)}$, the more consistently site i co-clusters with its siblings than with any rival type in view V ; the lower the margin, the more it co-clusters with a rival type than with its siblings.

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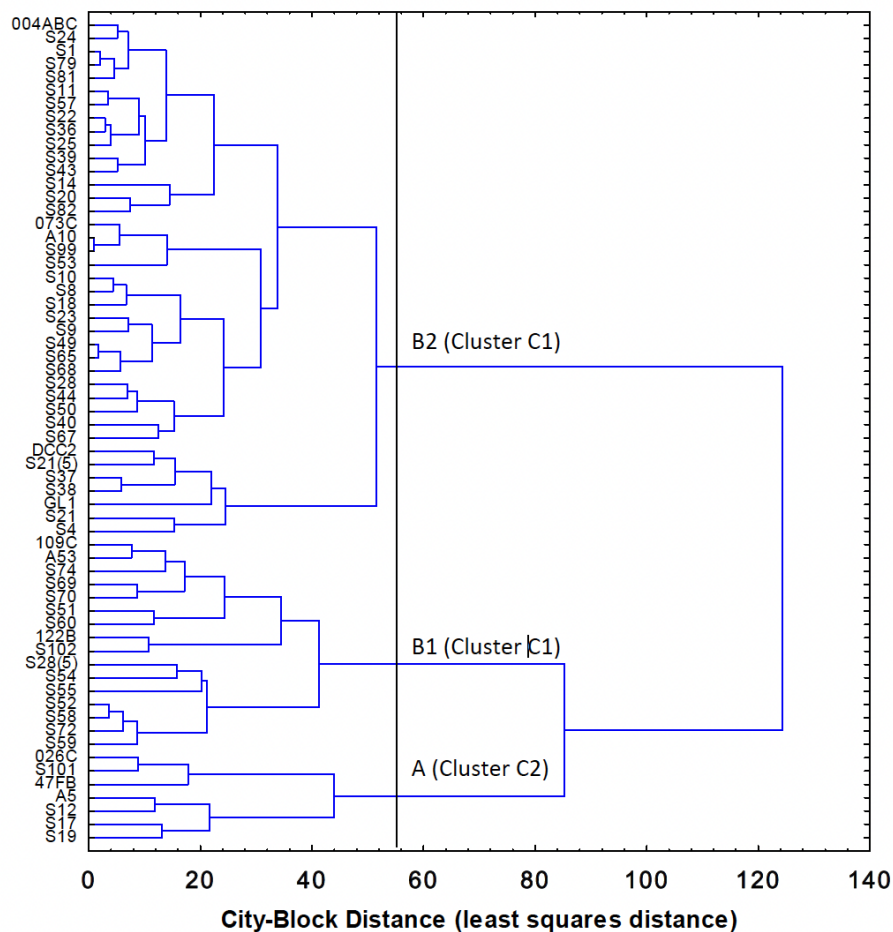


Figure 2 Dendrogram of REF sites ($n = 62$) grouped according to similar zoobenthic community composition in the 1991, 1999 and 2004/5 Lake Huron-Lake Erie Corridor analysis (Ward's method clustering city-block distances of octave-transformed relative abundances of zoobenthic taxa).

Figure 2.8: Branch-selective definition of community types in the 2008 reference dendrogram. The tree first separates B_2 from $\{B_1, A\}$, and only at a lower height splits B_1 from A . Cluster C_1 is assigned to B_2 and B_1 jointly, and Cluster C_2 to A alone—a non-monophyletic grouping that no single horizontal cut can produce, since A is topologically closer to B_1 than B_2 is.

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Table 2.5. Summary of observed number of Lake Huron-Lake Erie Corridor sites in each cluster (columns) identified by zoobenthic taxa relative abundances and membership predicted (rows) by discriminant function classification (Appendix III) on the basis of habitat characteristics measured at those sites

Group	% Correct	Observed	
		Cluster C1	Cluster C2
Cluster C1	98	54	1
Cluster C2	71	2	5
Total	95	56	6

Figure 2.9: The confusion matrix of a LDA classifier trained in the least-contaminated sites with their environmental descriptors and the 2008 branch-selective types. The unbalanced distribution of types (more C_1 than C_2) and extremely high accuracy is caused by subjectively assigning labels in a hindsight manner to achieve environmental separability.